Azure Optimization Tool

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The Azure Optimization Tool is an extensible solution designed to generate optimization recommendations for Azure environment, also it collects data for the resources which are unused and underutilized for a long period of time.

What you can get

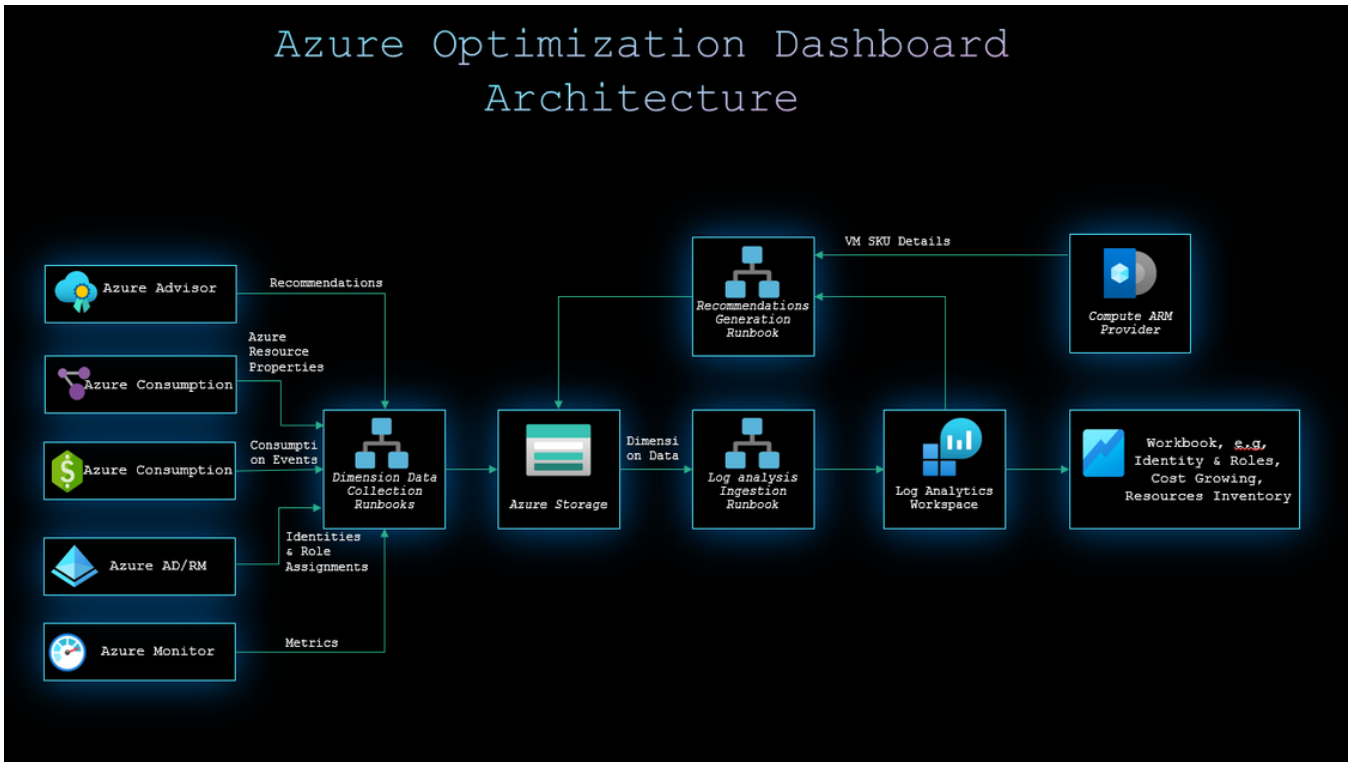
This tool will give you different workbook with all Azure optimization opportunities, coming from both Azure Advisor and from custom recommendations included in the engine. Recommendations are updated every 7 days.

Built-in custom recommendations

Besides collecting **all Azure Advisor recommendations**, Azure optimization tool includes other custom recommendations, such as:

- Cost
    - Augmented Advisor Cost VM right-size recommendations, with fit score based on Virtual Machine guest OS metrics (collected by Log Analytics agents) and Azure properties
    - Underutilized VM Scale Sets
    - Unattached disks
    - Standard Load Balancers without backend pool
    - Application Gateways without backend pool
    - VMs deallocated since a long time ago (forgotten VMs)
    - Orphaned Public IPs
    - Underutilized Azure SQL Databases (DTU-based SKUs only)
    - Storage Accounts without retention policy in place
    - Underutilized App Service Plans
    - App Service Plans without any application
    - Premium SSD disk has been underutilized
- High Availability
    - Virtual Machine high availability (availability zones count, availability set, managed disks, storage account distribution when using unmanaged disks)
    - VM Scale Set high availability (availability zones count, managed disks)
    - Availability Sets structure (fault/update domains count)
- Performance
    - VM Scale Sets constrained by lack of compute resources
    - SQL Databases constrained by lack of resources (DTU-based SKUs only)
    - App Service Plans constrained by lack of compute resources
- Security
    - Service Principal credentials/certificates without expiration date
    - NSG rules referring to empty or inexisting subnets
    - NSG rules referring to orphan or removed NICs
    - NSG rules referring to orphan or removed Public IPs
- Operational Excellence
    - Load Balancers without backend pool
    - Service Principal credentials/certificates expired or about to expire
    - Subscriptions close to the maximum limit of RBAC assignments
    - Management Groups close to the maximum limit of RBAC assignments
    - Subscriptions close to the maximum limit of resource groups
    - Subnets with low free IP space
    - Subnets with too much IP space wasted
    - Empty subnets
    - Orphaned NICs

Architecture:



Dimension Data Collection Runbooks periodically collect data from Azure Advisor (Cost recommendations) and Azure Resource Graph (Virtual Machine and Managed Disk properties) and dump to an Azure Storage Account container a selection of properties as CSV raw data . The Azure Storage Account repository has a long-term retention and can be used as well as a source for replaying transformation and load operations. We are using here only a couple of data sources, but others could easily be plugged in (Azure Billing, Azure Monitor metrics, etc.).

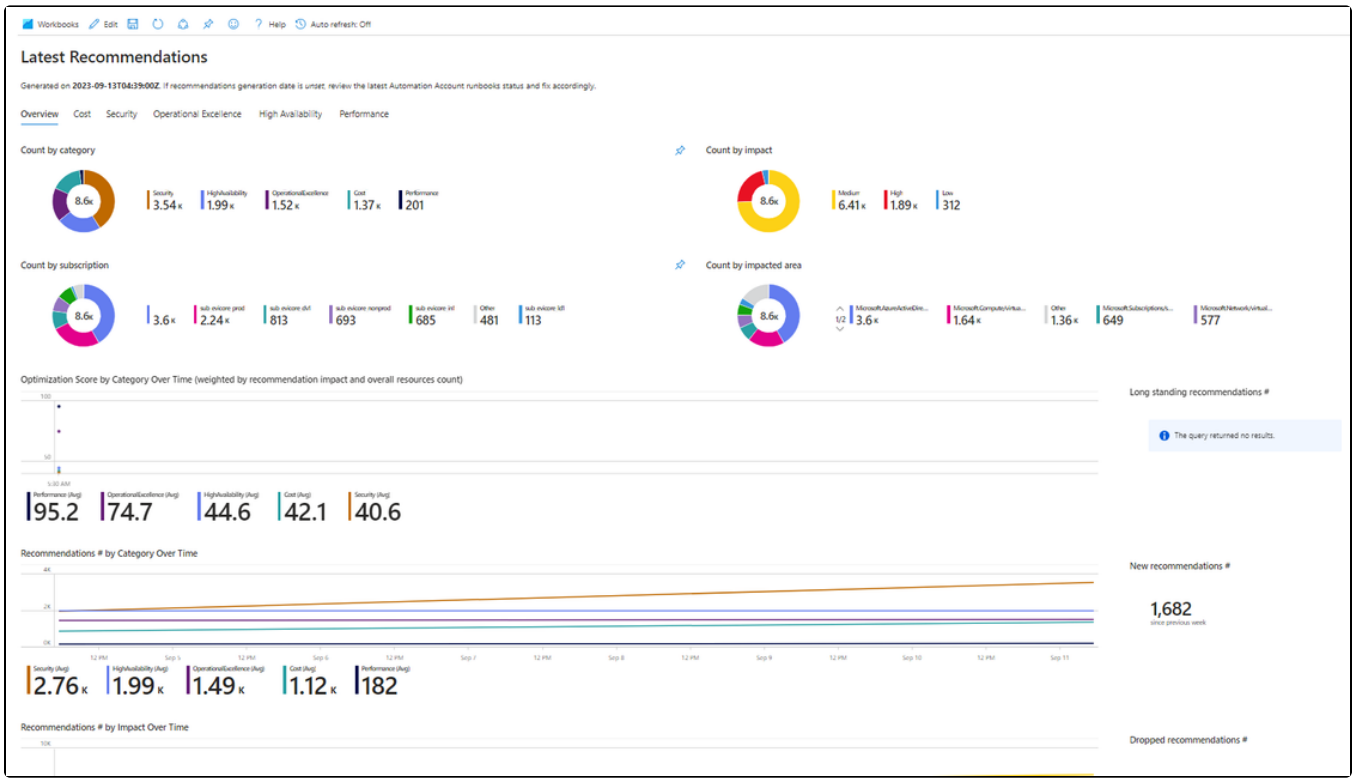
Raw data files are ingested into the Log Analytics workspace by the **Log Analytics Ingestion Runbook**, with all the data together in the same repository and with the power of Log Analytics, it is possible to augment Azure Advisor recommendations with very useful information and even generate new recommendation types out of Azure Resource Graph and performance counters .

Recommendation Generation Runbooks have recommendation type-specific logic that queries the Log Analytics workspace as well as other sources of information (e.g., the Compute ARM provider) to augment or generate a recommendation. In this scenario, Advisor Cost recommendations are merged with VM properties and performance metrics aggregates. Other scenarios could easily arise from the available data, such as recommendations for deleting orphaned disks. Again, all these runbooks dump the augmented recommendations to an Azure Storage Account container.

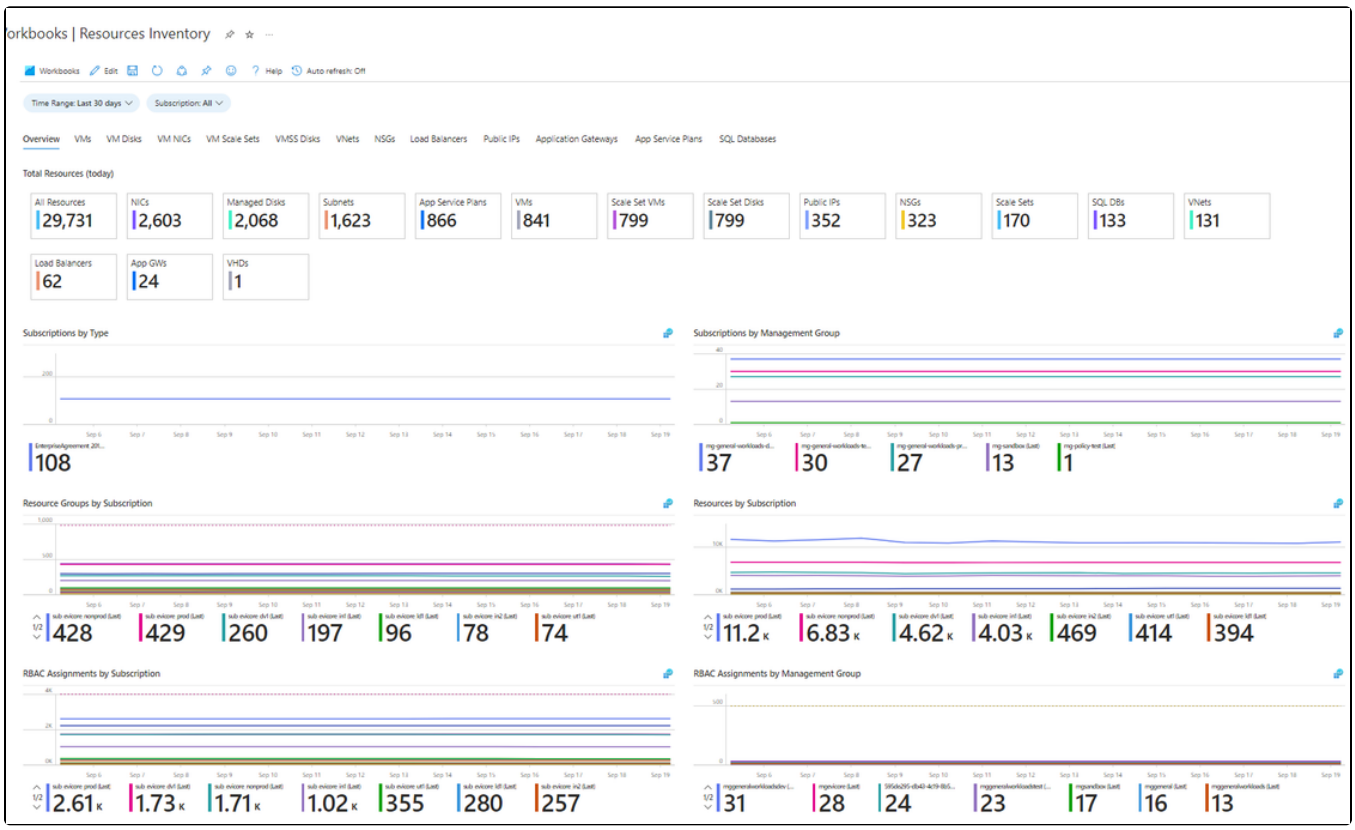
Finally, the SQL Server Injection Runbook periodically parses the raw recommendation files and sends them as new rows to an Azure SQL database containing all the recommendations history. The result of the process is visible in a Power BI dashboard – see sample view below.

Below are the workbook links:

Recommendation Workbook



Resources Inventory Workbook



Identities and Roles workbook

orkbooks | Identities and Roles

Workbooks | Edit | Help | Auto refresh: Off

Azure AD Tenant: All

For full functionality, this workbook requires Azure AD exports. If you had not assigned the Global Reader role to the AOE managed identity yet, please do so (see requirements).

Azure AD objects count (today)

Group	User	ServicePrincipal	Application
9.21k	8.37k	6.17k	2.47k

Azure AD users count (today)

Member	Guest
8.29k	75

Azure AD roles count (today)

Directory Readers	Other	Application Developer	Application Administrator	Security Administrator	Global Reader	Directory Writers
20	14	7	7	5	5	3

Azure RM roles count (today)

Contributor	Other	Reader	Virtual Machine Administr...	Network Contributor	Log Analytics Contributor	Storage Blob Data Contrib...
2.76k	2.42k	1.05k	627	529	401	277

Azure AD Credentials | Azure AD Roles | Azure RM Roles | Azure Classic Roles

Credentials about to expire | Credentials not set to expire | Credentials expired

Expires in (days): 30

Application	Expires On	Key Type	Tenant ID	Application ID	Role Assignments
eviCoreDev-HI^2 Project-a0c6645e-c3da-4a78-9ef6-04a...	9/29/2023, 2:16:00 AM	Password	595de295-db43-4c19-8b50-183dfd4a3d06	8138ffaf-3d71-4e92-983c-9f434866f487	⚠ 2
eviCoreDev-XPlatform-5fab10df-31da-4b2f-a1a8-4e2f51...	10/8/2023, 12:40:00 AM	Password	595de295-db43-4c19-8b50-183dfd4a3d06	99fe897d-dcf6-433e-8667-dd775c5f8b3e	⚠ 2
eviCoreDev-XPlatform-a0c6645e-c3da-4a78-9ef6-04ab6a...	9/24/2023, 1:59:00 AM	Password	595de295-db43-4c19-8b50-183dfd4a3d06	a39f0550-99fb-4940-bbe6-b0bf08605916	⚠ 1
eviCoreDev-ImageOneDomainApps-ed7f55f8-8333-41d1...	10/14/2023, 2:24:00 AM	Password	595de295-db43-4c19-8b50-183dfd4a3d06	5653b108-a64b-4d75-b2e5-58d225676907	⚠ 1
eviCoreDev-XPlatform-5fab10df-31da-4b2f-a1a8-4e2f51...	9/22/2023, 8:53:00 PM	Password	595de295-db43-4c19-8b50-183dfd4a3d06	32ef87b1-51fe-4a8d-8fbf-7c1778e85d92	0
troybraman-troybraman-fa99a186-8ff8-4e9c-bfb7-7bc37...	9/30/2023, 1:06:00 AM	Password	595de295-db43-4c19-8b50-183dfd4a3d06	b40776d4-1644-4e4e-935d-90e5f216c61b	0

Roles assigned to application (selected above)

This query could not run because some parameters are not set.
Please set: selectedApp